

DAVID ABABIO AWUNI

Language-Model Behaviour | Reliability of Model Evaluation

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RESEARCH INTERESTS

Language-model behaviour; reliability of model evaluation; LLM-as-judge systems; evaluator bias; prompt sensitivity; multilingual evaluation around African-language content.

EDUCATION

Bachelor of Science in Computer Science, First Class Honours

Jan 2021 - Aug 2024

Kwame Nkrumah University of Science and Technology (KNUST)

Kumasi, Ghana

Undergraduate thesis: A Multi-Layered Hybrid Deep Learning Framework for Cyber-Physical Intrusion Detection in Climate-Monitoring IoT Systems | Supervisor: Dr. Linda Banning

Selected coursework: Artificial Intelligence, Machine Learning, Computer Security, Data Communications, Network Programming, Database Systems, Software Engineering

SELECTED RESEARCH EXPERIENCE

Independent Researcher — Family-Conditioned Preference in LLM-as-Judge Evaluation

2026 - Present

Independent research project

Ghana

- Designed a fully crossed evaluator-provenance audit using Llama 3.1, Qwen 2.5, Gemma 2, and Yi 1.5 as candidate and judge families across 194 validated prompts from MT-Bench, AlpacaEval, and WildBench.
- Built a reproducible pairwise-evaluation pipeline producing 9,312 ordered judgments and 4,656 AB/BA-reconciled comparisons; developed the **Family Preference Score (FPS)**, a signed matrix contrast for measuring same-family support.
- Implemented prompt-clustered bootstrap and permutation tests, Bradley-Terry quality controls, quasi-binomial GEE, placebo-label shuffling, leave-one-family-out checks, and a 72-specification multiverse analysis.
- Completed the audit, blinded human calibration, and float16 replication; manuscript available as a preprint, with a reproducibility archive prepared for release.

Independent Researcher — AfriSyCo: Assertive Framing and Verification

2026 - Present

Independent multilingual NLP research project

Ghana

- Evaluated seven open-weight checkpoints on 100 source questions in Hausa, Igbo, Kinyarwanda, Swahili, Twi, and Yoruba, yielding 1,415 turn-1-correct model-language-item observations.
- Used native-language follow-ups and a controlled 2×2 experiment with English follow-up framing to distinguish assertion and verification effects.
- Estimated effects of 30.4 percentage points for assertion, -17.4 for verification, and +19.7 for their interaction in the controlled experiment; tested wording and checkpoint sensitivity.
- Documented substantial wording and checkpoint heterogeneity, showing why prompt realization must be treated as part of the measurement instrument.

Undergraduate Researcher — Hybrid Intrusion Detection for Climate-Monitoring IoT

Jan 2024 - Aug 2024

Department of Computer Science, KNUST

Kumasi, Ghana

- Developed a multi-layer defense-in-depth framework combining environmental-sensor telemetry with network-traffic classification for cyber-physical intrusion detection.
- Trained and evaluated sensor-level and CNN-BiLSTM classifiers on ToN_IoT and CICIoT2023, including binary detection and 25-class attack classification.
- Applied SHAP-based explainability to examine feature importance and support transparent interpretation of security-model decisions.

MANUSCRIPTS AND PRESENTATIONS

Awuni, D. A., Achenie, L. E. K., Partey, B. T., Owusu, E. G., & Sowah, N.-N. D. (2026). Who Judges Matters: Measuring Family-Conditioned Preference in LLM-as-Judge Panels. Preprint.

Awuni, D. A., Gyening, R.-M. O. M., & Owusu, E. G. (2026). AfriSyCo: Measuring Assertive Framing, Verification, and Wording Sensitivity Around African-Language Content. Preprint.

Awuni, D. A., Gyening, R.-M. O. M., & Twum, F. (2024). *Enhancing Student Engagement in Online Lecture Streaming Platforms through Real-Time Quizzes*. Paper presented at the E-Learning International Conference 2024, KNUST.

ACADEMIC AND PROFESSIONAL EXPERIENCE

Research and Teaching Assistant <i>Department of Computer Science, KNUST</i> - Supported laboratory instruction for 150+ undergraduate students in Computer Security and Data Communications, including programming guidance, technical troubleshooting, assessment, and course support. - Provided structured feedback on 12 sections of graduate students' research drafts, focusing on methodological clarity and evidence alignment. - Mentored 20+ undergraduate students on thesis design, research methodology, technical implementation, and presentation.	Oct 2024 - Aug 2025 <i>Kumasi, Ghana</i>
Frontend Developer Intern <i>Codextra, Accra, Ghana</i> Developed web and mobile application features, reviewed pull requests, and presented weekly technical progress reports across concurrent software projects.	Aug 2023 - Dec 2023

SELECTED TECHNICAL PROJECT

Edustream — AI-Enhanced Live Lecture Platform <i>KNUST student research and software project</i> - Co-developed desktop, web, and mobile components for live lecture delivery and real-time student engagement. - Adapted an NLP question-generation pipeline to generate contextual quizzes from lecture content; the project produced a conference presentation at E-Learning International Conference 2024.	Jan 2024 - Sep 2024 <i>Kumasi, Ghana</i>
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HONOURS AND AWARDS

Vice-Chancellor's Students' Excellence Award — First Runner-Up, All-Rounded Exemplary Student	2024
Fifty-50 Club Scholarship — Merit- and need-based undergraduate scholarship	2021 - 2024

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL, Dart
Machine Learning: PyTorch, TensorFlow, Hugging Face, Scikit-learn, PEFT/LoRA, CNNs, LSTMs, transformer-based models
NLP and Evaluation: Multilingual benchmarking, LLM-as-judge evaluation, prompt design, human annotation, inter-annotator agreement, language-ID and tokenization analysis
Statistics: Bootstrap inference, permutation testing, GLMMs, GEE, Bradley-Terry models, effect sizes, multiple-comparison correction
Tools: Git, Docker, Linux, Google Colab, AWS, LaTeX, pandas, NumPy
Languages: Twi/Akan (native), English (fluent)

SELECTED LEADERSHIP AND SERVICE

Senate Speaker <i>Science Students' Association, College of Science, KNUST</i> Presided over 24 Senate sessions involving more than 50 student representatives and led review of an annual student-association budget exceeding GHS 100,000; organized an educational visit to the Parliament of Ghana for 30 student leaders.	Aug 2023 - Aug 2024 <i>Kumasi, Ghana</i>
Academic Committee Head <i>Computer Science Society, KNUST</i> Coordinated peer-tutoring initiatives and coached departmental teams for academic competitions, including a third-place finish in the Prof. Amekudzi College Quiz.	Aug 2022 - Sep 2023 <i>Kumasi, Ghana</i>
Organizer and Lead Facilitator — Programming and Skills Boot Camp <i>Student-led community program</i> Designed and delivered practical training in Python, web development, and digital literacy for 80+ students.	Oct 2023 - Nov 2023 <i>Kumasi, Ghana</i>

References available upon request.